Dhruv Kapri

GeoAI | GIS & Spatial Analysis | LiDAR & Remote Sensing | Machine Learning | Urban Analytics
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Education

Indian Institute of Technology, Kharagpur — West Bengal, India
Bachelor of Architecture: 8.31/10 — Dec 2021 – Apr 2026

Publications

- *Integrating AI-Driven Personas and Procedural Visualization for Complete Communities Design and Urban Planning for Large-Scale Urban Development*, **HCII 2025, Springer Nature Switzerland**
- *A New Contrastive Learning-based Vision Transformer for Sentiment Analysis using Scene Text Images*, **International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI 2024)**

Projects

GIS-Driven Rooftop Solar Potential, Washington DC (SIH × ISRO) — *Live* | *GitHub*

- Built a city-scale **Python** pipeline: **LiDAR DSM**, **GRASS r.sun** radiation, **RANSAC** roof planes, PVWatts yield
- Benchmarked on Glover Park, DC: specific yield about **1%** off **Esri**'s tutorial (1162 vs 1150); 12-day radiation sample
- Scored **100,064** DC rooftops in 66 tiles; census-tract (TIGER/ACS, DOE LEAD energy burden) **equity** choropleth
- Shipped a public **MapLibre + PMTiles** web map; resumable per-tile GeoParquet cache, 165 tests, open-source
- Earlier **Rhino-Grasshopper + Ladybug** prototype: LOD-1 model, full-year roof & facade radiation & shading

Research Experience

OCAD University, Canada | *Mitacs Globalink Research Intern* — May 2024 – Aug 2024

- Developed geospatial data pipelines to validate **61 amenities** across **3 districts** for urban-planning spatial modeling
- Applied travel-pattern analytics & unsupervised clustering on **158 case studies** to identify urban layout insights
- Built a persona-driven simulation framework using **112 LLM-based personas**; co-authored the **HCII 2025** paper

Universiti Malaya, Malaysia | *Research Fellow* — Dec 2022 – Jan 2024

- Developed a **Vision Transformer** with contrastive learning: **98.58%** accuracy, **+5.8%** over CNN; **IJPRAI** paper
- Built the data-augmentation & contrastive pre-training pipeline, improving feature separability & generalization
- Collected & analyzed a **12K+ image** dataset across positive, neutral & negative sentiments with **95% agreement**

National University of Singapore, Singapore | *BIM Intern* — Nov 2024 – Feb 2025

- Curated and classified **50+ architectural plans** across typologies and time periods to build structured BIM datasets
- Built **Revit BIM models** and automated material take-offs using Dynamo scripts for accurate quantity extraction
- Analyzed quantities against **Singapore government datasets** to identify long-term material utilization trends

Industry Experience

SLnko Energy Pvt. Ltd., India | *Graduate Trainee - Product Engineer* — May 2026 – Present

- Building a **floorplan automation engine** in **Python**: **CP-SAT** lays out rooms & columns as bylaw-checked plans
- **Site intake**: **PostGIS** point-in-polygon zoning and seismic lookup applies **20** bylaw rules to buildable plot polygons
- Built a **Tender Intelligence RAG app** (Docling + hybrid retrieval) cutting solar-EPC review from days to hours

Veritas Group, Indonesia | *Digital Solutions Intern* — Feb 2025 – Jul 2025

- Built internal analytics & ML-assisted dashboards delivering live project insights, cutting stakeholder queries by **30%**
- Shipped Python-based parametric & automation tools for Rhino + Grasshopper, cutting design-iteration time by **40%**

Technical Skills

GIS & Spatial: ArcGIS Pro | GRASS GIS | PostGIS | LiDAR | Raster & Vector | GeoParquet | MapLibre
Languages: Python | SQL | C++ | JavaScript | TypeScript
ML & DL: PyTorch | TensorFlow | scikit-learn | Pandas | NumPy | Transformers | ViT | OpenCV
Tools: Git | Docker | FastAPI | GitHub Actions | Rhino | Grasshopper | Ladybug | Revit